

Module name: **Respiratory system (Year II Anesthesia Students)**

Module ECTS: 3 ECTS

Common infections in the Respiratory System

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December, 2023

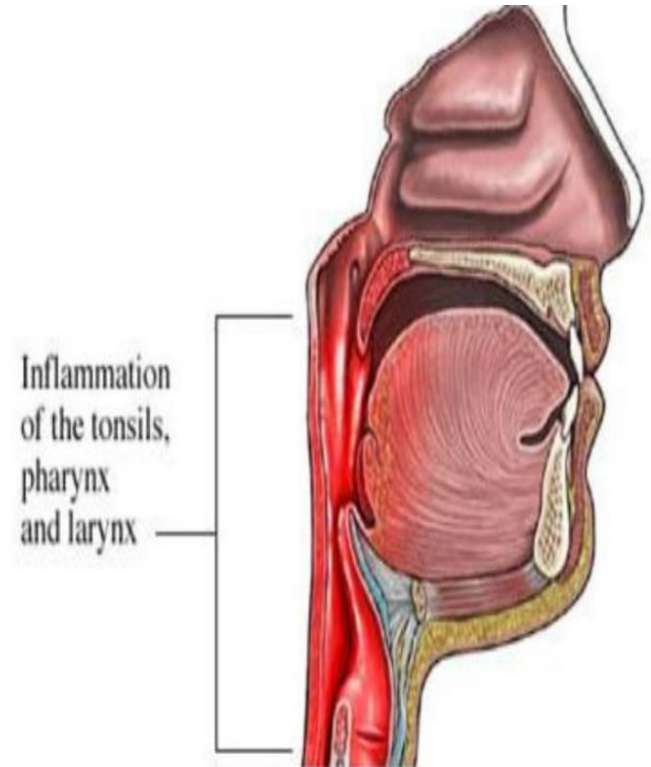
Lecture Outline

Selected (common) infections involving the respiratory tract

- Pharyngitis (sore throat)
- Diphtheria
- Pertussis
- Epiglottitis
- Pneumonia
- Common cold
- Corona virus diseases
- Influenza
- Measles

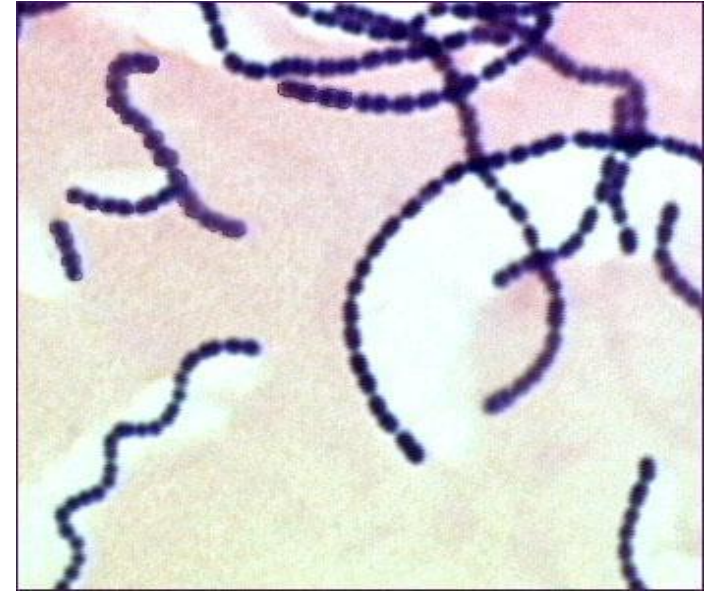
Pharyngitis

- Inflammation of the pharynx
 - Tonsils are also most often affected
- Up to 85% are caused by viruses
 - Usually self-limiting
- Potentially dangerous form is caused by bacteria
 - **Group A- Beta hemolytic streptococci (GA β HS)**



Group A- Beta haemolytic streptococci (GA β HS)

- Gram-positive cocci in chains
- Usually capsulated
- Facultative anaerobes
- Catalase-negative
- Streptococci are classified based on:
 - **Cell wall carbohydrate antigen** (Lancefield classification)- Rebecca Lancefield
 - A, B and D (more frequent)
 - ✓ *Streptococcus pyogenes* (Group A)
 - Non-groupable streptococci
 - ✓ *S. pneumoniae*
 - **Hemolysis on blood agar**
 - B-hemolytic
 - *S. pyogenes*
 - *S. agalactiae*
 - **α -hemolysis**
 - **γ -hemolysis**



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GABHS

Source of infection and transmission

- *S. pyogenes* is a normal flora of the skin and mucous membranes of humans

Nearly 5% of all people, regardless of age, carry the organism in their throats

- Baseline streptococcal pharyngeal carriage rate is roughly 5%
- Reaches **15% to 50%** in school-age children in temperate climates during epidemics of GAS pharyngitis.

- **Transmission occurs via aerosolized droplets**

- Person-to-person transmission via droplets or secretions
- Food and water-borne outbreaks occur

GABHS pharyngitis

Important virulence factors

- Structural components
 - M protein- adhesion factor
- Toxin production
 - Streptolysins (S and O types)
- Enzyme production
 - Streptokinase
 - Hyaluronidase

GABHS pharyngitis

Clinical manifestation

- Reddened pharynx with exudates, cervical lymphadenopathy
- Also called “**strep throat**”



Figure 13-2. Inflammation of the oropharynx (soft palate) and tonsils in a patient with *Streptococcus pyogenes* pharyngitis. Notice the redness on the oropharynx and tonsils. (Note: Viral pharyngitis can appear similar to this image.) Image courtesy of the Centers for Disease Control and Prevention.

- **Post streptococcal complications**

- Acute rheumatic fever (ARF)**

- Non-suppurative inflammatory **lesions of heart**, joints, subcutaneous tissues, & CNS
 - Follows an upper respiratory GAS infection
 - Molecular mimicry?: Ig's to M protein react with cardiac myosin, synovium, and articular cartilage
 - Occurs in **0.4 to 3%** of untreated GAS pharyngitis cases

- **Acute glomerulonephritis**

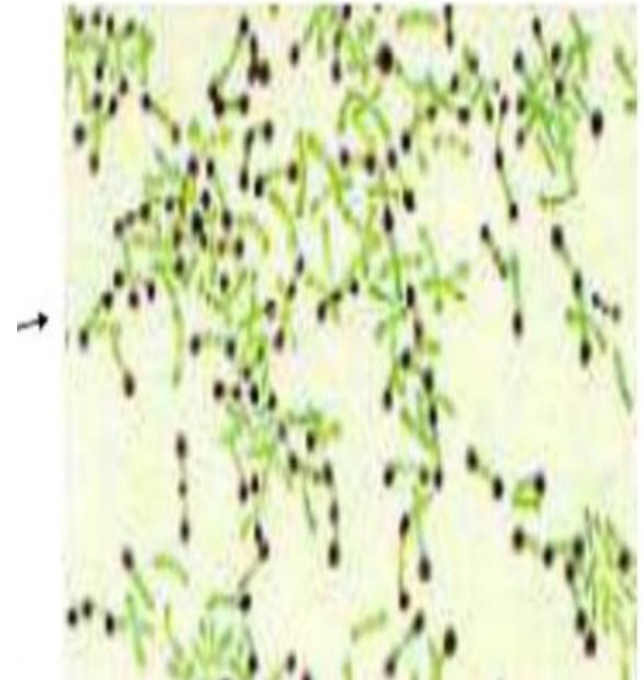
- Believed to be immunologic etiology: cross-reaction vs. deposition of Ab-Ag complexes

GABHS pharyngitis

- **Laboratory diagnosis**
 - Antigen detection
 - Culture throat swab
 - Serology- antistreptolysin O (ASO)
- **Treatment**
 - Penicillin
 - Erythromycin
- **Penicillin prophylaxis** with long-acting preparations is used to prevent recurrences of ARF during the most susceptible ages (5 to 15 years)

Respiratory diphtheria

- An acute, bacterial disease of respiratory tract
 - Caused by toxin-producing strains of *Corynebacterium diphtheriae*
 - Gram-positive, facultative anaerobes
 - Irregular shaped resembling Chinese letters



Respiratory diphtheria

- **Source of infection and transmission**
 - Normal flora of upper respiratory tract
 - Infection sources include infected persons and carriers (rare)
 - Transmitted through droplets
 - Also sometimes indirectly via contaminated objects
- **Virulence factors**
 - Diphtheria toxin

Respiratory diphtheria

Clinical manifestation

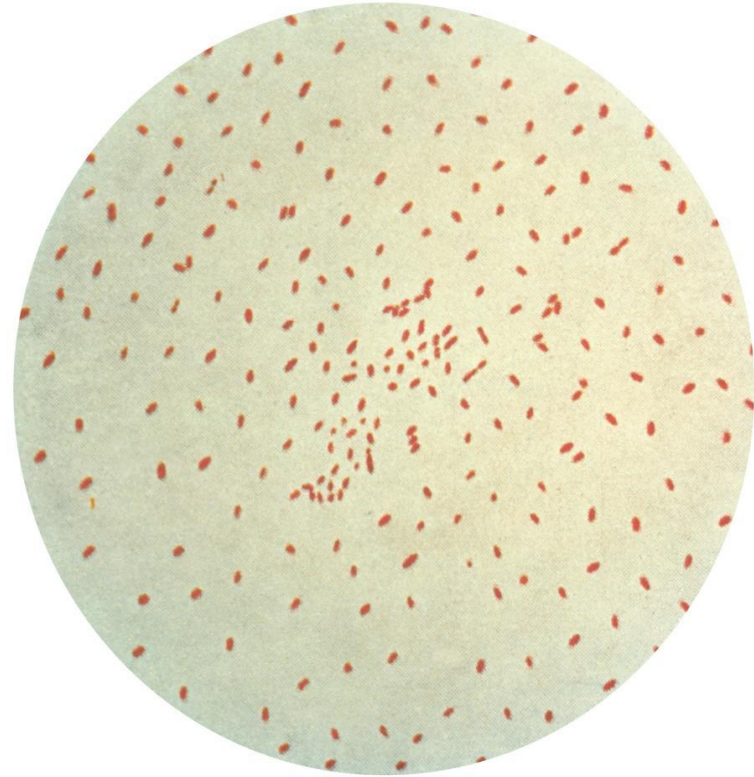
- Acute diseases (IP 2-6 days)
- Characterized by a sore throat and low-grade fever
 - Produces obstructive **pseudo-membranes** in the upper respiratory tract (**croup**)
- Mainly a disease of small children in developing countries
 - Common in ages of 6–8 years



- **Laboratory diagnosis**
 - Microscopy
 - Culture
 - Toxin detection
- **Treatment**
 - Anti toxin
 - Penicillin & cephalosporins
 - Erythromycin
- **Prevention**
 - Immunization with **diphtheria toxoid (DPT)**

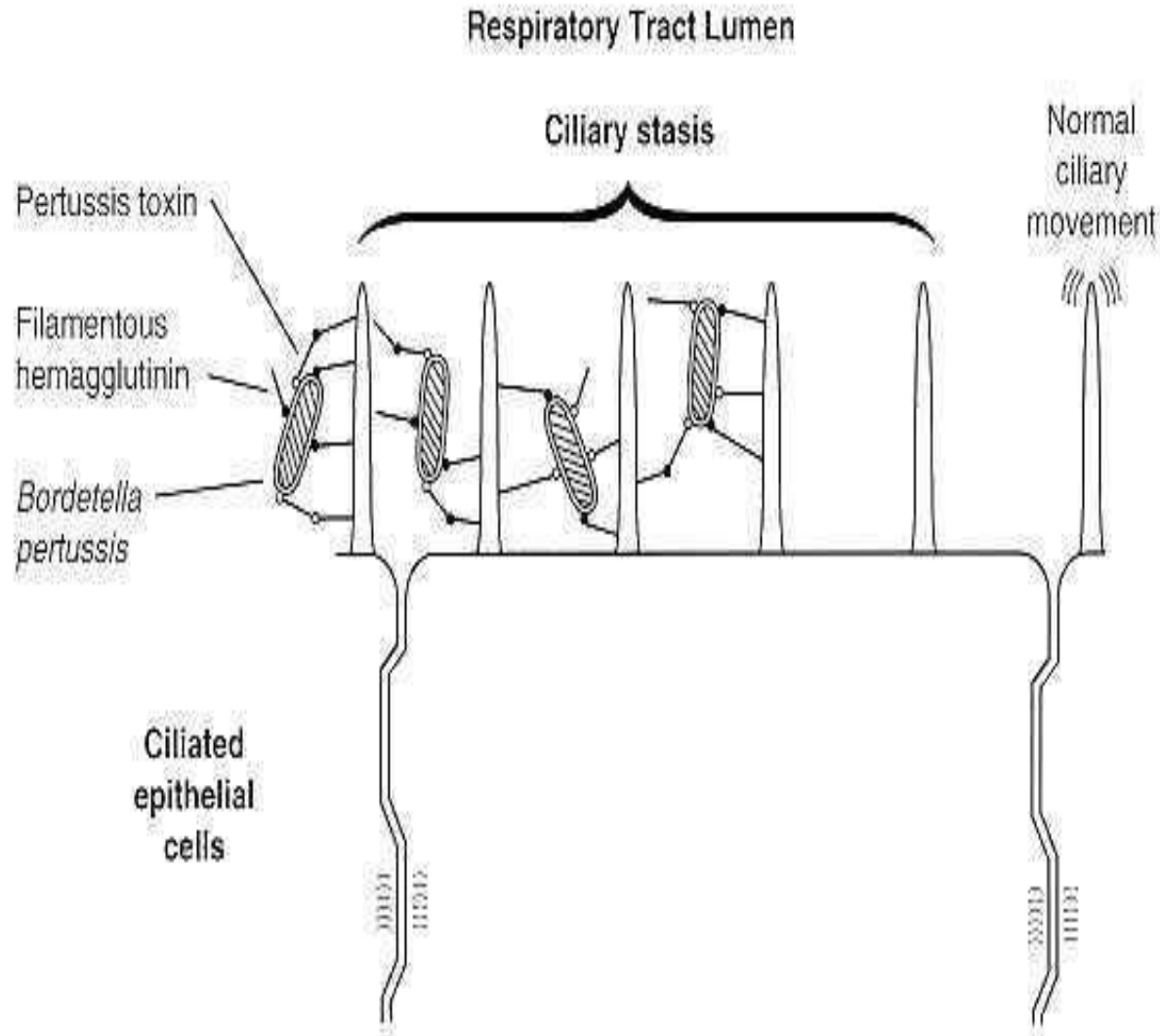
Pertussis/whooping cough

- A highly contagious respiratory disease caused by ***Bordetella pertussis***
 - Gram-negative
 - Aerobic
 - Encapsulated coccobacillus



- **Source of infection and transmission**
 - Person to person through droplets
 - No carrier stage
 - Can survive in the environment
 - Common in children under age of 5 years
- **Virulence factors**
 - Pili-involved in adherence
 - Pertussis toxin
 - Tracheal cytotoxin
 - Lipopolysaccharide (LPS)

Synergy between pertussis toxin and FHA in binding to ciliated respiratory epithelial cells



- **Clinical manifestation**

Three stages:

- **Catarrhal phase**

- Mild upper RTI, develops after an average IP of 7 to 10 days, indistinguishable from common cold
- Highly infectious but not very ill

- **Paroxysmal phase**

- Occurs after IP of 1 to 2 weeks
- Characterized by explosive cough/whooping cough
- Breathing problems (cyanosis, and convulsions)

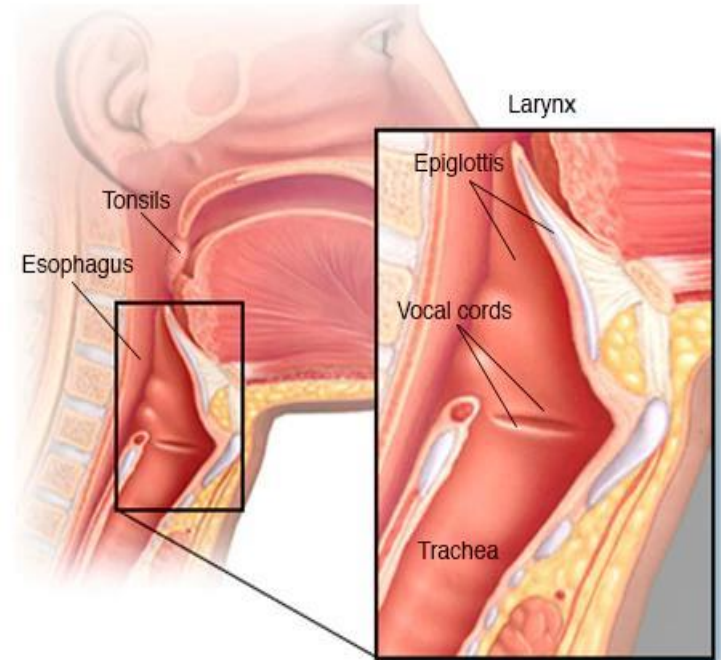
- **Convalescent stage**

- Less severe cough that may persist for several months

- **Laboratory diagnosis**
 - Culture
 - Serology
- **Treatment**
 - Erythromycin
- **Prevention**
 - Pertussis vaccine in the form of DTP

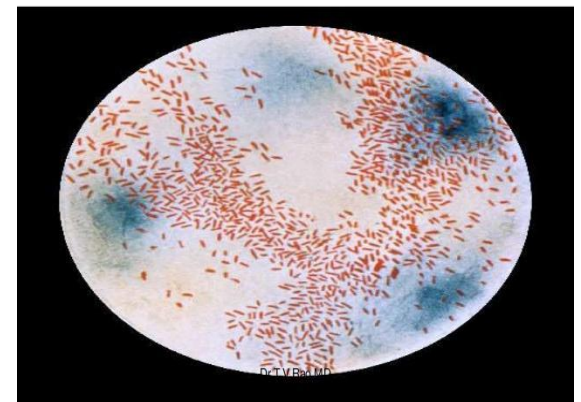
Epiglottitis

- A potentially life-threatening condition of the epiglottis
- Can be caused by viruses, burns from hot liquids, direct injury
- Commonly caused by bacteria-
***Haemophilus influenzae* Subtype b (Hib)**
 - Gram negative coccobacillus
 - Fastidious
 - Encapsulated
 - Facultative anaerobes
- **Hib also causes**
 - Meningitis
 - Bacteremia
 - Endocarditis



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Haemophilus influenza

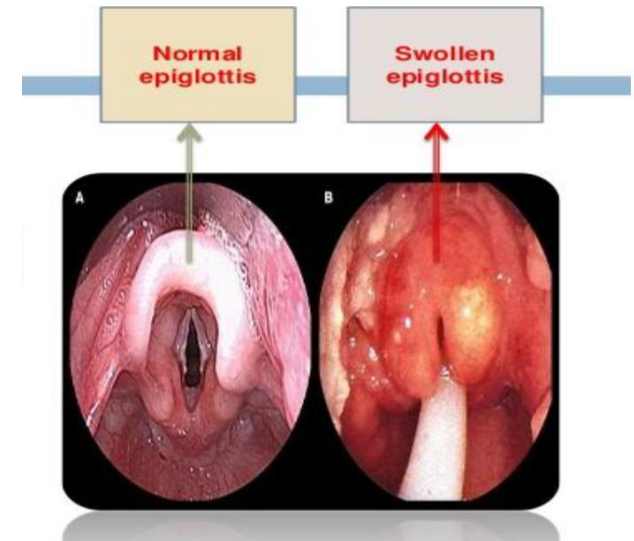


Hib Epiglottitis

- **Source of infection & transmission**
 - Through direct contact with respiratory droplets from nasopharyngeal carrier or case patient
 - Common in children < 12 months of age (unvaccinated ones)
- **Virulence factors**
 - Capsule-most important
 - Pili
 - IgA proteases

Hib Epiglottitis

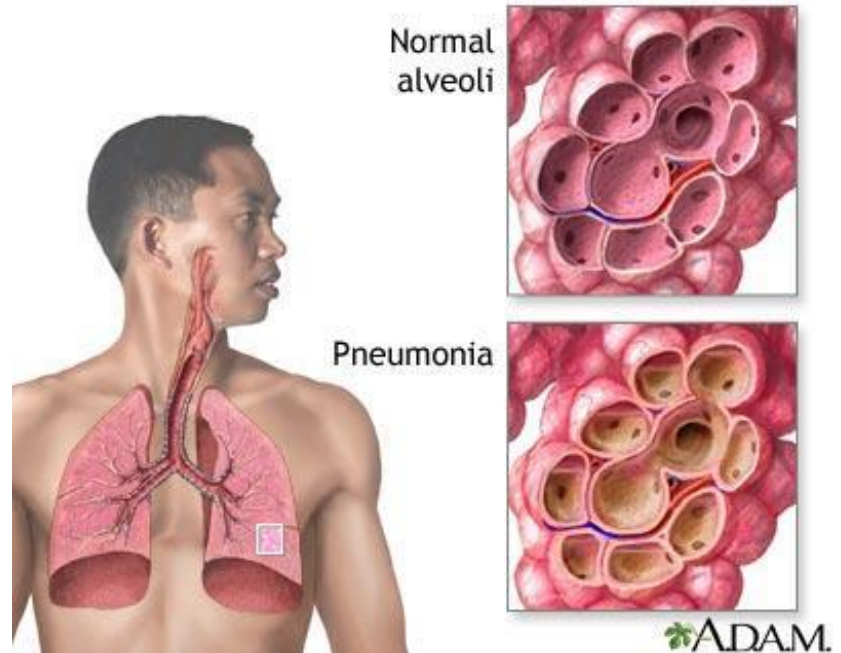
- **Clinical manifestation**
 - Swelling of the epiglottitis
 - Inability to swallow secretions
 - Hoarseness



- **Laboratory diagnosis**
 - Capsular polysaccharide antigen detection
 - Culture of nasopharyngeal swabs
 - Grow on chocolate agar
- **Treatment**
 - third-generation cephalosporins
- **Prevention**
 - Hib conjugate vaccine

Pneumonia

- Infection that inflames the air sacs in one or both lungs
- Can be caused by
 - Bacteria
 - Viruses
 - Fungi
 - Toxic chemicals



Pneumonia

Bacterial pneumonia

- **Pneumococcal pneumonia**
 - *Streptococcus pneumoniae*
- **Staphylococcal pneumonia**
 - *Staphylococcus aureus*
- **Influenzal pneumonia**
 - *Haemophilus influenzae* (non capsulated strains)
- **Gram-negative bacterial pneumonia**
 - *Klebsiella pneumoniae*
- **Mycoplasma pneumonia**
 - Mycoplasma

- **Viral pneumonia**

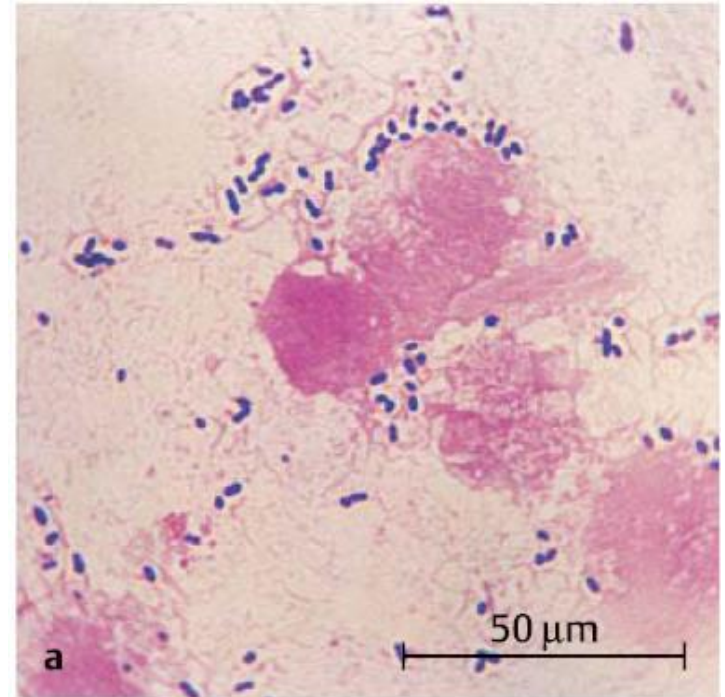
- E.g. due to Influenza A virus

- **Fungal pneumonia**

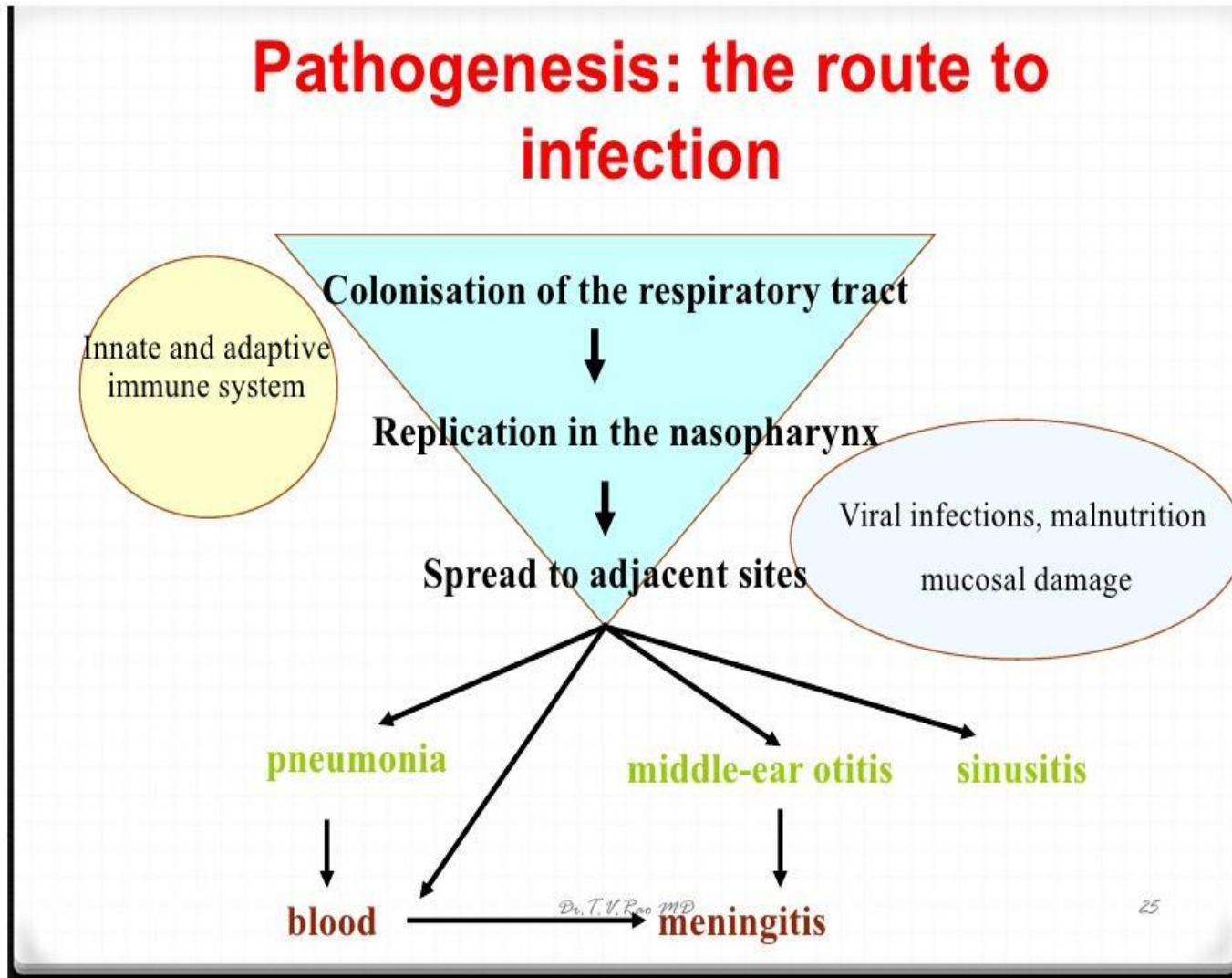
- *Aspergillus* species
- *Histoplasma capsulatum*
- *Cryptococcus neoformans*

Pneumococcal pneumonia

- Caused by *Streptococcus pneumoniae*
 - Gram-positive, **lancet-shaped** cocci
 - Seen as diplococci or in short chains
 - α hemolytic colonies
- **Encapsulated**
 - Over 90 different **serovars** based on the chemical structure of the capsule polysaccharides



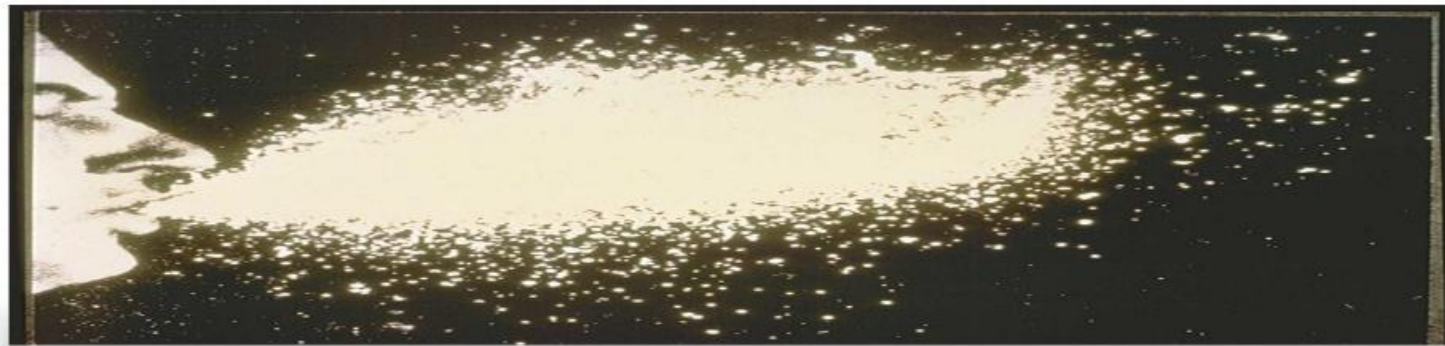
Source of infection & transmission



Source of infection & transmission

Pneumococcus: colonisation and transmission

- Exclusively a human pathogen, part of the normal microbial flora of the upper respiratory tract
- Transmission: by droplet secretions
- Temporal pattern: winter-early spring
- Communicability: Unknown. Probably as long as organism in respiratory secretions



PNEUMOCOCCAL CARRIER STATE

- Disease occurs in persons who are already asymptomatic carriers
 - Carrier rates 38%-60% in preschool children
 - 29%-35% in school children
 - 9%-25% in junior high school students
 - 18%-29% in adults with children at home
 - 6% in adults with no children at home
 - Virtually all children <2 years of age become carriers

Pneumococcal pneumonia

- **Virulence factors**
 - **Capsule**
 - Pneumolysin
 - Secretory IgA (sIgA) protease

- **NEUMOCOCCAL DISEASE: PNEUMONIA a major Complication**
 - Bacteraemia in 15–30% of patients with pneumonia
 - **High mortality despite appropriate antibiotic therapy**
 - Overall case fatality rate 15–20% for pneumococcal bacteraemia
 - Higher case fatality rates (30–40%) for elderly persons

- **Laboratory Diagnosis**

- Capsule Swelling Tests (Quellung reaction) for serotyping
- **Culture:** Blood, CSF, Sputum and others

- **Treatment**

- Penicillin G is the drug of choice
- Erythromycin, vancomycin
- Third-generation cephalosporins

- **Prevention**

- Pneumococcal vaccines
 - Old 23-valent Pneumococcal Polysaccharide Vaccine (PPV23)
 - New 7-valent Pneumococcal Conjugated Vaccine (PCV7)

FUNGAL PNEUMONIA

- Fungal pneumonia is a type of lung infection caused by FUNGI
- **Common in immuno-compromised people**
 - HIV infection important risk factors
- Symptoms of fungal pneumonia are similar to bacterial or viral pneumonia
- Case mortality in fungal pneumonias can be as high as **90%** in immunocompromised patients

Fungal pneumonia

Common causes

- **Coccidioidomycosis**:- begins with an often self-limited respiratory infection
- **Pneumocystis pneumonia**:- typically occurs in immunocompromised people, especially AIDS.
- **Cryptococcosis** :-contracted through inhalation of soil contaminated with the yeast, it can manifest as a pulmonary infection and as a disseminated one
- **Aspergillosis**: - ubiquitous molds found in soil, grains, and in decomposing organic materials
 - Common in HIV patients
- **Histoplasmosis**: Found in soil, often associated with decaying bat guano or bird droppings.

- **Laboratory diagnosis – Fungal pneumonia**
 - Microscopic examination
 - 10% KOH preparations
 - Indian ink staining (for cryptococcosis)
 - Fungal culture
- **Treatment**
 - a range of antifungal drugs (Fluconazole , Amphotericin, Caspofungin, etc)
- **Prevention**
 - avoid exposure

Tuberculosis (TB)

- An infectious disease caused by *M. tuberculosis*
- Generally affects the lungs-**Pulmonary TB**
- Any organ can be affected- **Extrapulmonary tuberculosis (EPTB)**
- **Genus Mycobacteria** is important
 - Rod shaped, obligate aerobic bacteria
 - Lipid-rich cell walls (mycolic acids)
 - Acid fast bacteria
 - Several species
 - *M. tuberculosis*- TB in humans
 - *M. bovis*- TB in animals
 - *M. avium*- causes a TB-like disease in AIDS patients
 - *M. leprae*- causes leprosy

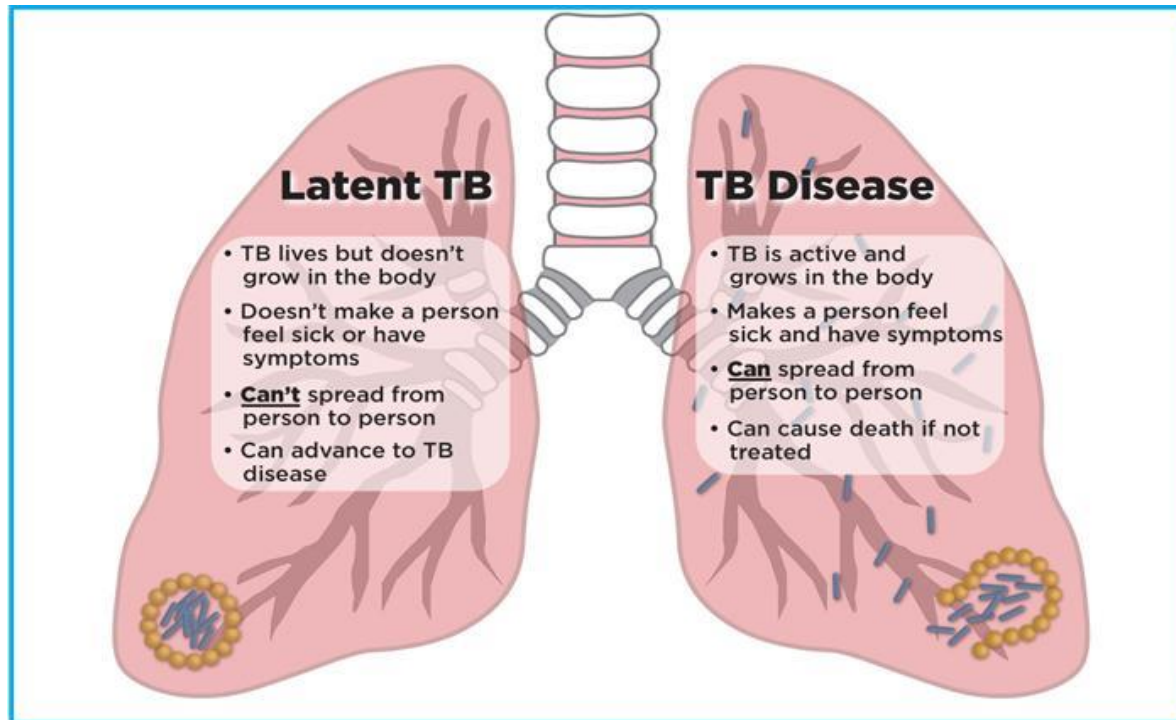
Transmission-pulmonary TB

- **Via droplet nuclei** generated during talking, coughing, and sneezing
 - Droplet nuclei contains > 3 bacilli.
- Sneezing generates the most droplet nuclei by far
 - Can spread to individuals up to 10 feet away
- **Risk factors**
 - Close contact with large populations of people
 - schools, nursing homes, dormitories, prisons
 - Poor nutrition, alcoholism
 - HIV infection, the #1 predisposing factor
 - Progression from latent to active TB is much higher in HIV patients

- **Latent TB**

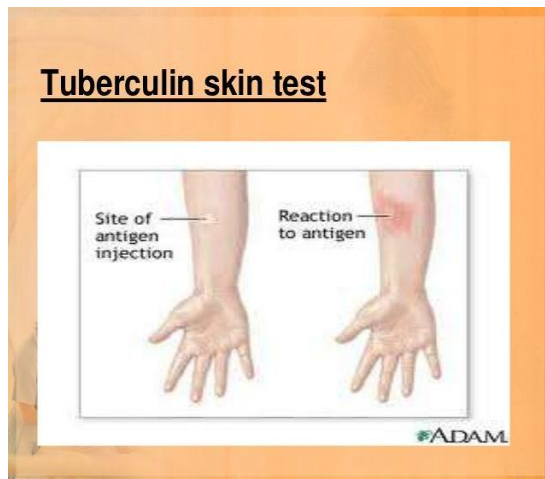
- *M.TB.* is in the body but the immune system is keeping the bacteria under control by macrophages
- No symptoms of TB
- Not infectious

Latent TB vs. TB diseases



- **Virulence factors**
 - Doesn't produce toxins
 - Intra-cellular multiplicaiton and cell damage

- **Laboratory diagnosis**
 - Microcopy
 - Culture
- **Latent TB**
 - Skin test (PPD)



— *Mycobacterium Tuberculosis* —

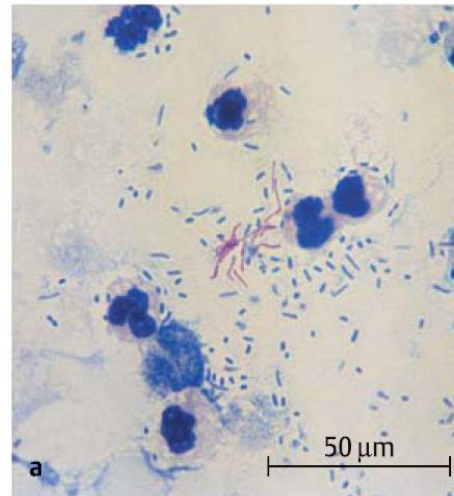


Fig.4.12 **a** Ziehl-Neelsen staining of a urine preparation: Fine, red, acid-fast rods, which tend to stick together. Clinical diagnosis: renal tuberculosis.

b Culture of *M. tuberculosis* on egg nutrient substrate according to Löwenstein-Jensen: after four weeks of incubation rough, yellowish, cauliflowerlike colonies.



PCR-based TB diagnostic test

- Gene Xpert MTB/RIF test
- Fast, sensitive, and automated
 - Used for active TB

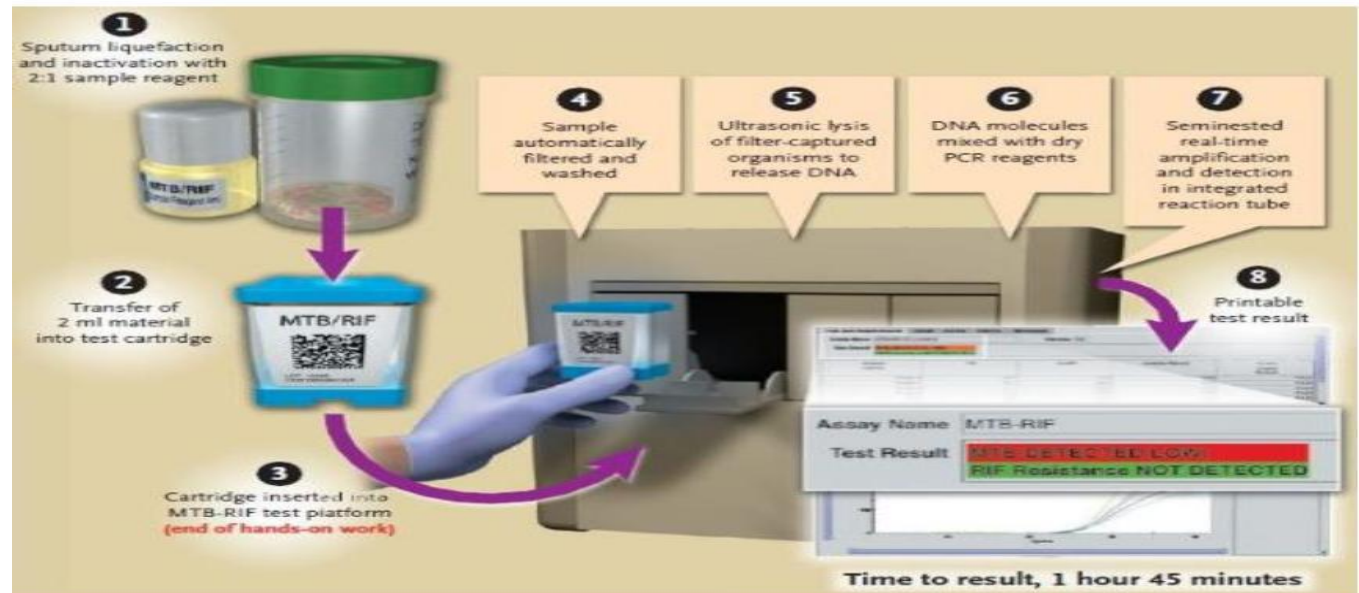
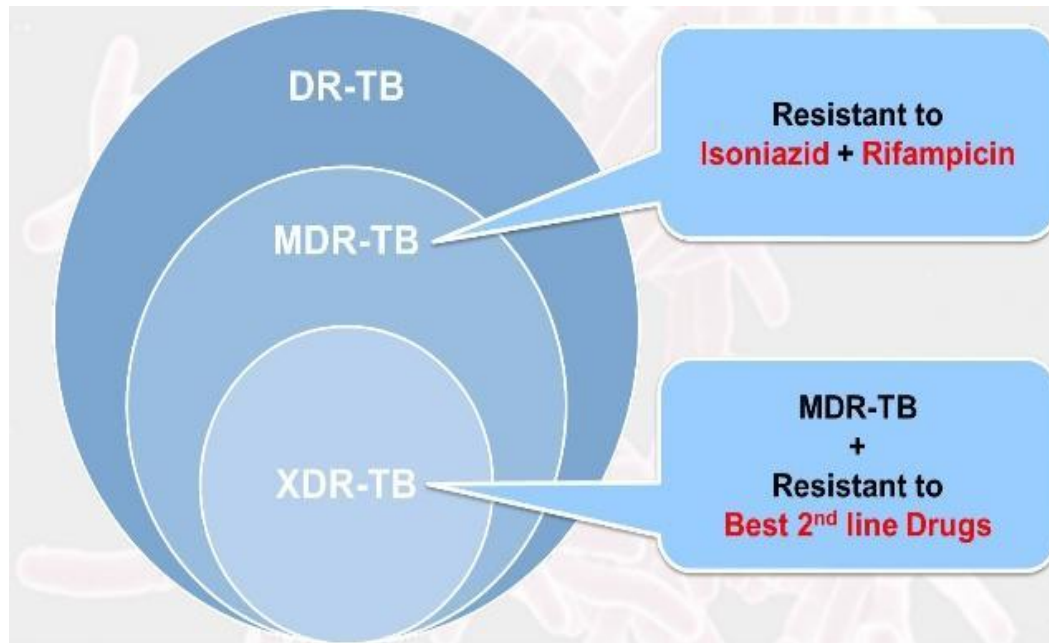


Fig. 11 PCR-based TB diagnostic test - Xpert MTB/RIF.

- **Treatment**
 - Combination of drugs used
 - First line (**new patients**) : rifampin (RIF), isoniazid (INH), pyrazinamide (PZA) and ethambutol (EMB)- for six months
 - Re-treatment patients: Streptomycin (SM) included
 - **Second line:** to treat MDR-TB
 - Levofloxacin, moxifloxacin, bedaquiline, delamanid, and linezolid, & pretomanid
 - Longer regimens: 18-20 months
 - Shorter treatment regimens: 9 months (on piloting)
 - In non compliant patients, directly observed therapy (DOT) is important.

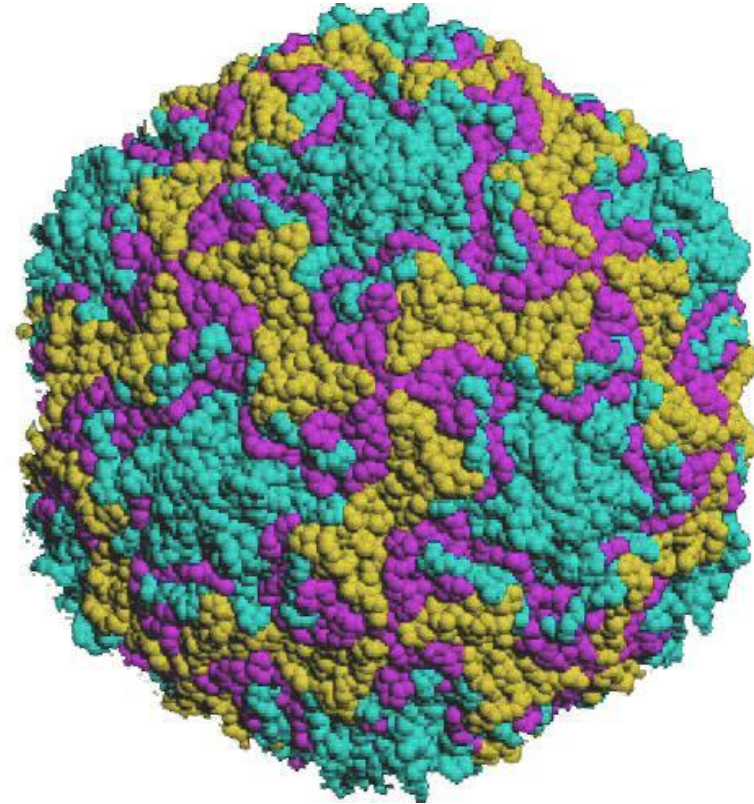
Drug resistant TB



- **Prevention and control**
 - Treatment of latent TB in high risk groups
 - BCG vaccination for children
 - Early diagnosis and treatment of active TB

Common cold

- Mainly caused by **rhinoviruses** of the family picornaviridae
 - RNA viruses with icosahedral symmetry
 - ss-RNA, + sense ,linear
 - Non enveloped
- More than 100 antigenic types
- **Thermo stable**- can survive for several days on surfaces



Rhinoviruses

Source of infection and transmission

- Two mechanisms
 - aerosols
 - Contact- major route
 - Direct- hand shakes
 - Indirect contact- contaminated inanimate objects

Common cold

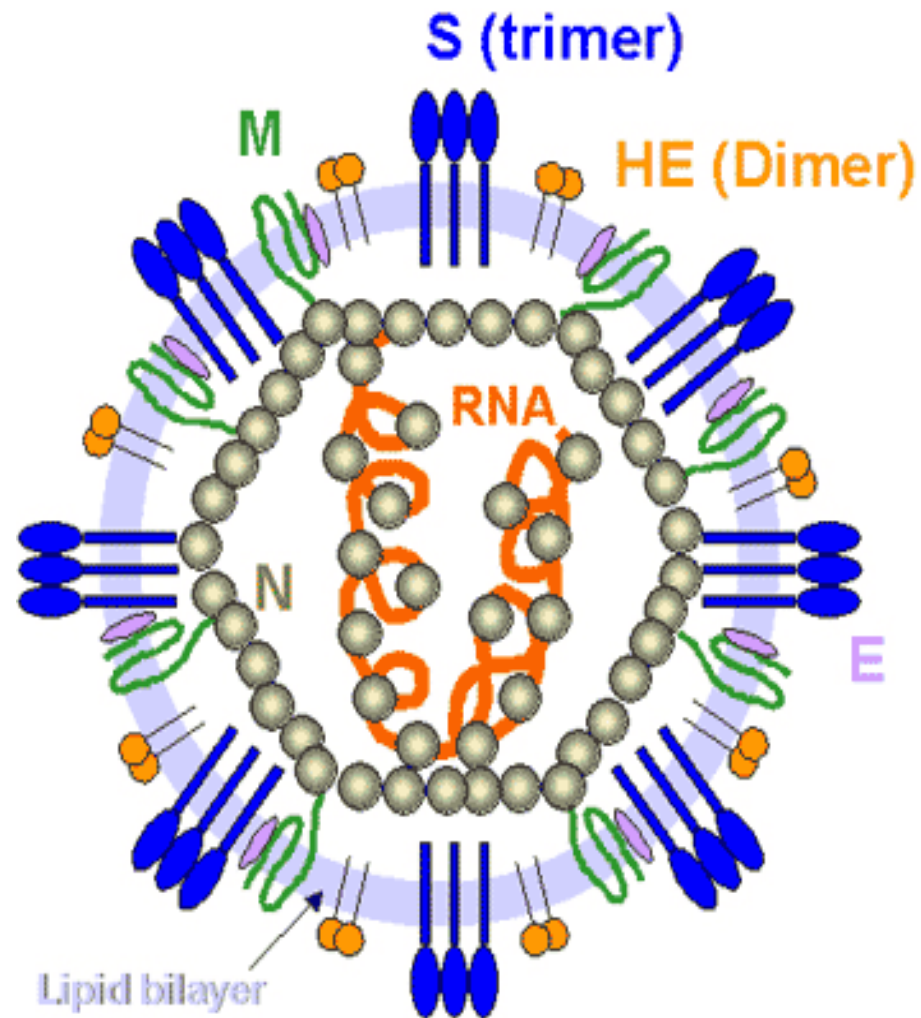
- **Clinical manifestation**
 - Mild URT illnesses
 - IP 2-4 days
 - Sneezing, nasal obstruction, nasal discharge, sore throat, cough, little/ no fever
 - Usually up to 7 days
 - 50% of rhinovirus infections are asymptomatic



- **Laboratory diagnosis**
 - Rhinovirus antigen using ELISA
 - Cell culture
- **Treatment**
 - No specific treatment
- **Prevention**
 - No vaccine
 - Hand hygiene

Coronaviruse diseases

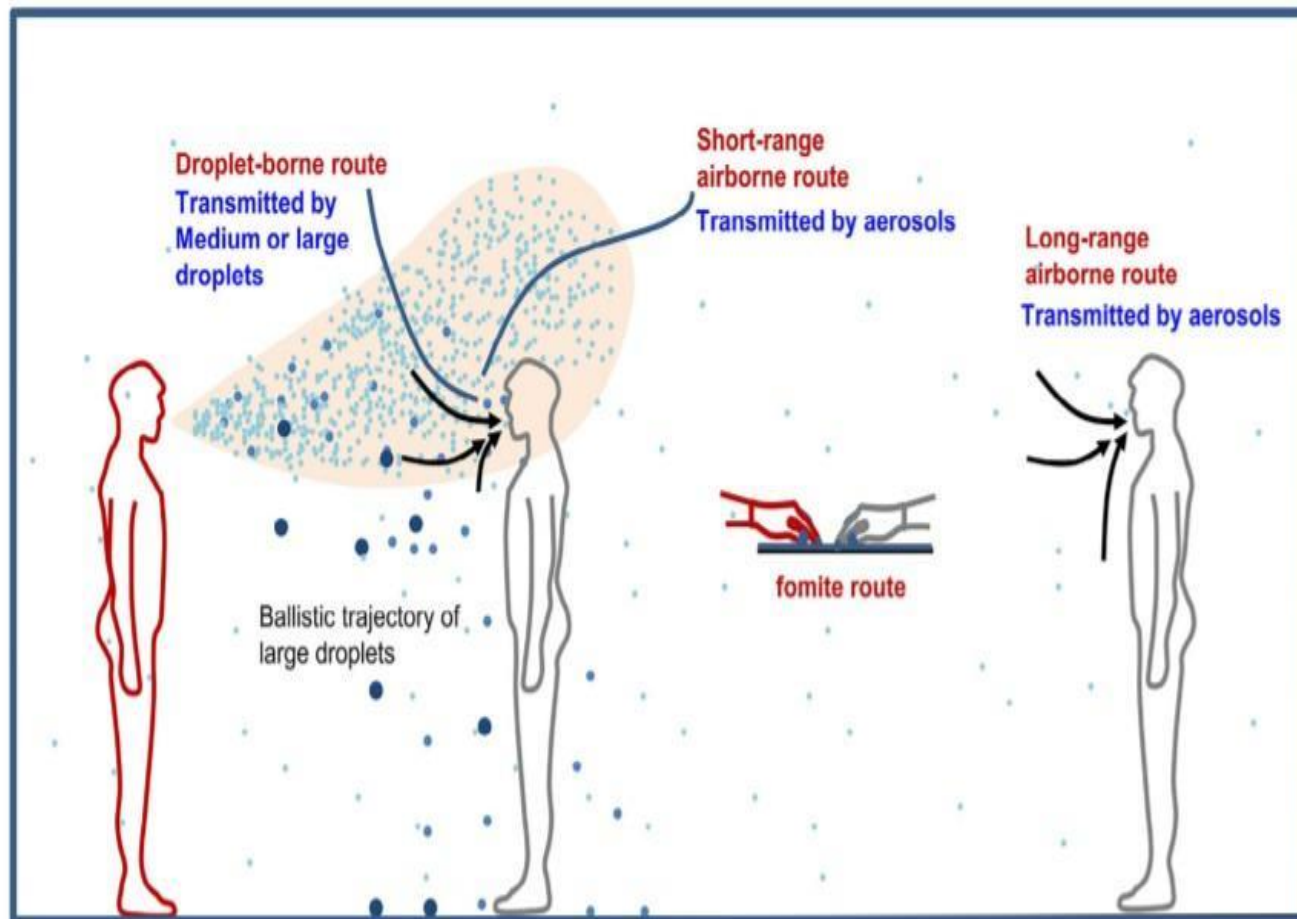
- Zoonotic infections caused by coronaviruses
 - Family coronaviridae
 - Genus coronavirus
 - Large, enveloped RNA viruses
 - Helical nucleocapsids
 - SS linear non segmented, +ve sense RNA



Outstanding characteristics

- High frequency of mutation
- High frequency of recombination during replication
 - May be a hybrid of mammalian and avian coronaviruses
- Known for causing outbreaks
 - about 15 species infecting animals, birds, & human
- Caused two important pandemics
 - SARS-CoV-1=> Southren China 2003
 - SARS-CoV-2 (COVID-19) -Wuhan city, China 2019

- **Source of infection and transmission**
 - droplet transmission (coughing or sneezing)
 - airborne transmission-medical settings
 - Contact with infected person (direct) or infected persons' secretions (indirect)
 - Fomite transmission- contaminated surfaces



- Large droplets ($>100\ \mu\text{m}$) : Fast deposition due to the domination of gravitational force
- Medium droplets between 5 and $100\ \mu\text{m}$
- Small droplets or droplet nuclei, or aerosols ($< 5\ \mu\text{m}$): Responsible for airborne transmission

Fig 4. Illustration of different transmission routes. Small droplets ($<5\ \mu\text{m}$), sometimes called aerosols, are responsible for the short-range airborne route, long-range airborne route, and indirect contact route; large droplets are responsible for the direct spray route and indirect contact route.

SARS-Cov-1 origin

- It is believed that virus was transmitted to humans from masked palm **civet** in food market of Southern China. It is possible that civet picked up the virus from a bird



SARS-Cov-2 origin

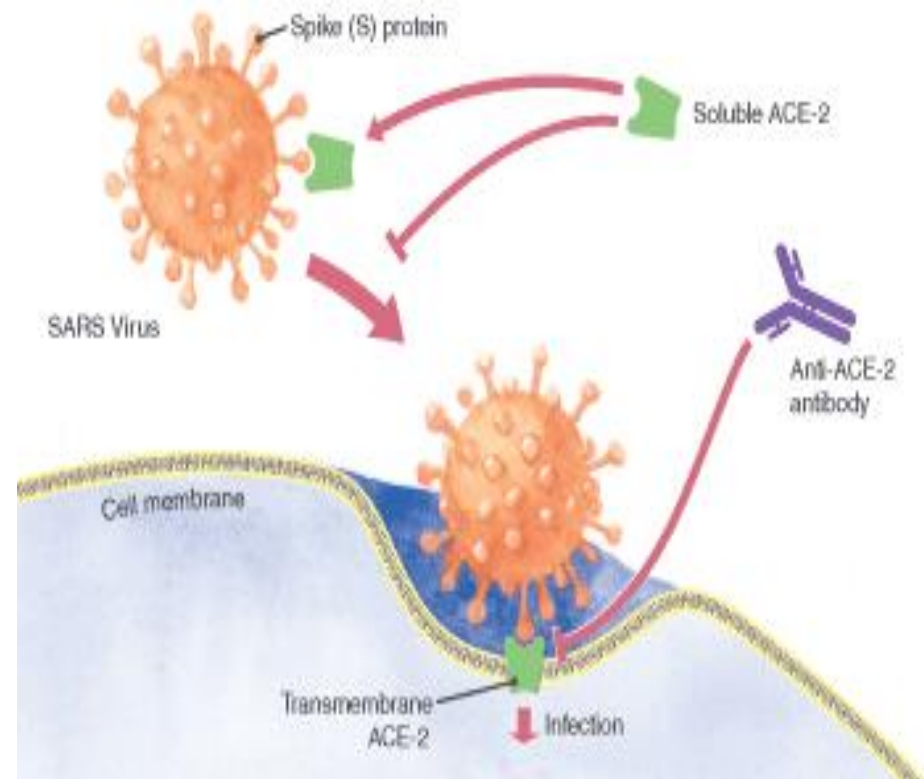
- Bat to a mammal (pangolin?) to human in Nov/Dec 2019
- Pangolins used in Chinese medicine
- Probable link to seafood/exotic animal market
- Other plausible theory:
 - Wuhan Level 4 Biohazard lab experimental animals sold for human consumption



Clinical manifestations

- Normal human coronaviruses cause 5-10% of common cold/URIs, with outbreaks to 30% of common cold
 - 229E and NL63 (alpha coronaviruses)
 - OC43 and HKU1 (beta coronaviruses)
- Novel coronaviruses predominantly in **LOWER respiratory tract**
 - SARS-CoV-1, SARS-CoV-2
 - Severe acute respiratory syndrome (SARS)
 - May involve the lungs (Pneumonia, shortness of breath)

Novel CoV attachment



- **ACE-2 Receptors**
 - Type 2 alveolar cells - highest
 - Bronchial epithelia
 - Tongue & buccal epithelia
 - Upper intestinal epithelia
 - Myocardial cells
 - Kidney proximal tubule cells
 - Bladder urothelial cells
- SARS-CoV-2 binds to ACE-2 Receptor 10-20x more strongly than SARS-CoV-1

Symptoms and Disease Course- SARS-CoV-2

- **Week 1:** Fever (77-98%) (intermittent or persistent), Fatigue/Malaise (11-52%), Dry cough (46-82%), dyspnea (3-31%);
 - Less common: Sputum (33%), Myalgia (15%), Headache (13%), Sore throat (14%), Diarrhea (4%), Nausea/Vomiting (5%), Nasal congestion (4%), Hemoptysis (1%)
- **Week 2** (~ day 6-9 of symptoms): ~ 15-20% develop severe dyspnea due to viral pneumonia
 - Hospitalization, supportive care, oxygen
- **Week 2-3:** Of hospitalized patients, 1/3 ultimately need ICU care, with up to half needing intubation (i.e. ~5% of total diagnosed cases need ICU)
 - Can rapidly decline (over 12-24 hrs) from mild hypoxia to frank Acute respiratory distress syndrome (ARDS)
 - Cytokine Storm, Multi-organ failure
 - Late stage sudden cardiomyopathy/viral myocarditis, cardiac shock

Cormorbidities and Risk Conditions

- Age
- Diabetes
- Coronary Heart Disease
- Hep B virus infection
- Cerebrovascular Disease
- COPD
- Cancer

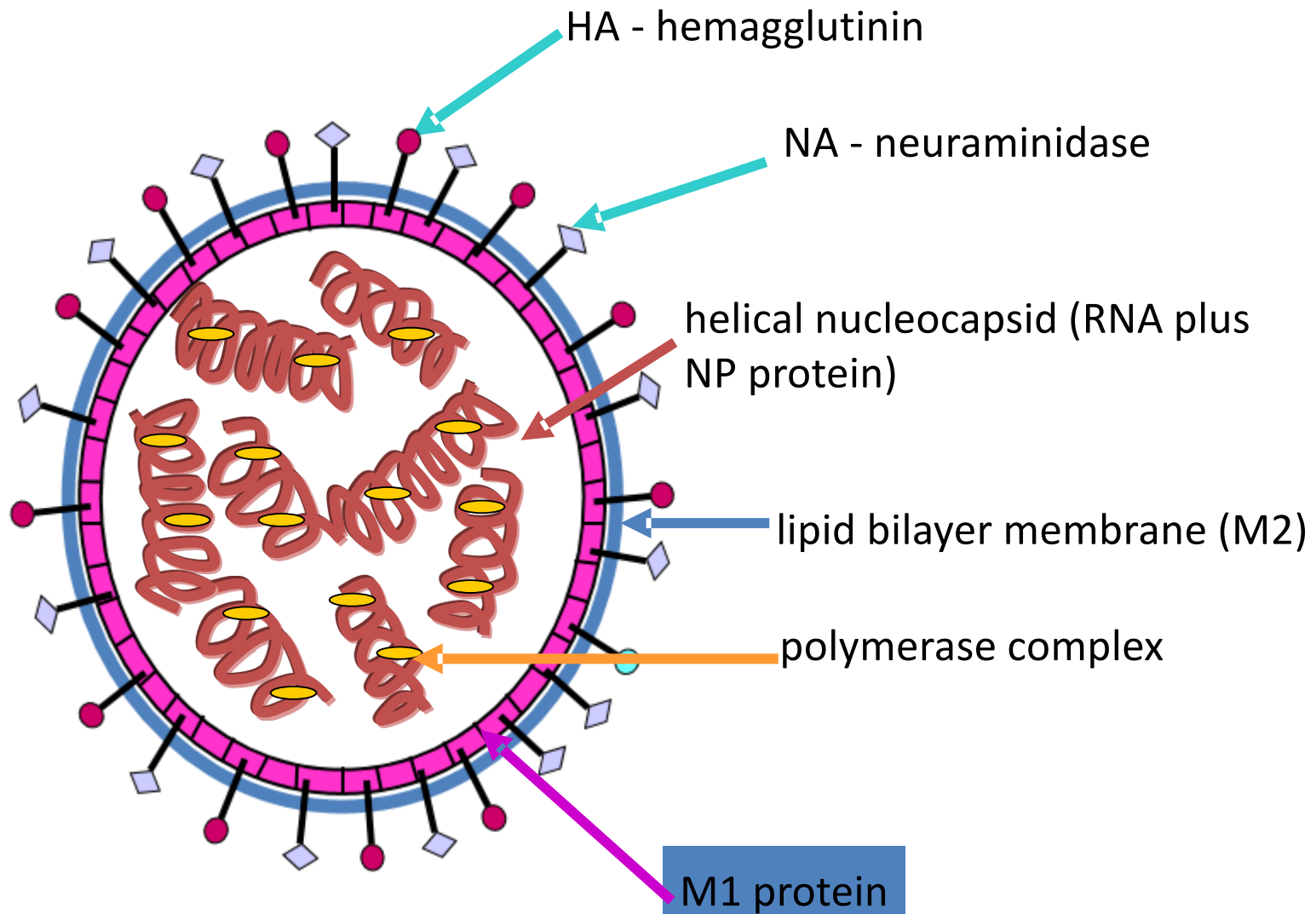
- **Laboratory diagnosis**

- Antigen detection- rapid test kits, ELISA
- NA detection- RT-PCR
- Cell culture- difficult to grow

- **Treatment- no cure**
 - Remdesivir
 - Anticoagulation drugs ("blood thinners")
- **Prevention and control**
 - Isolation of patients
 - Quarantine of those exposed
 - Use of barrier precautions
 - gloves
 - gowns
 - respirators by health workers
 - universal masking
 - Vaccines approved for emergency use
 - Pfizer
 - Moderna
 - AstraZeneca

Influenza

- Acute upper respiratory infection with systemic symptoms
- Caused by **Influenza viruses**
- Family Orthomyxoviridae
 - Helical nucleocapsid
 - Genome, segmented, –ss RNA
 - Enveloped, HA & NA



Type A, B, C : NP, M1 protein
sub-types: HA or NA protein

Three antigenic groups

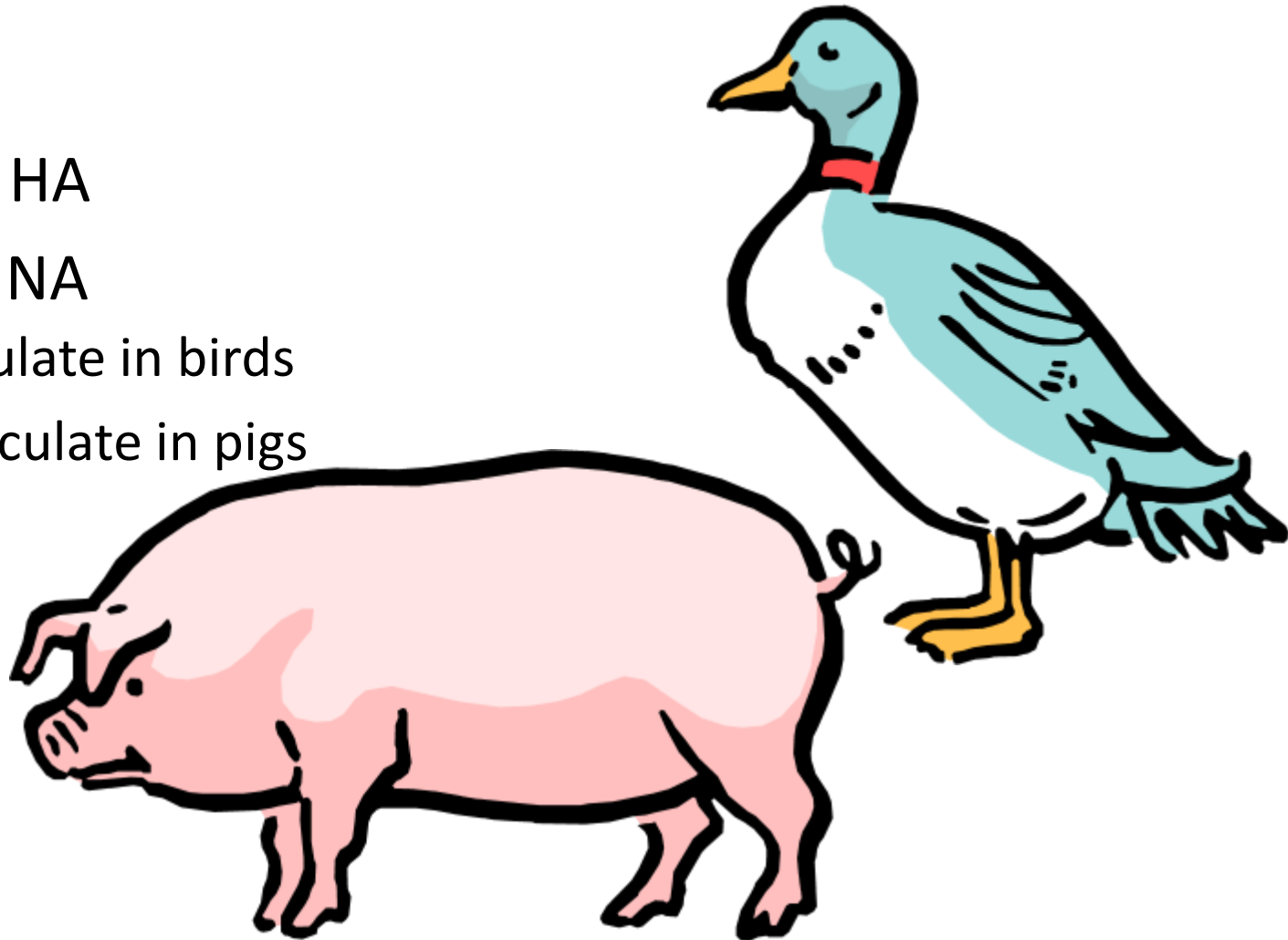
		A	B	C
	Severity	+++++	++	+
	Animal Reservoir	Yes	No	No
	Population spread	Pandemic, epidemic	Epidemic	Sporadic
	Antigenic changes	Shift, drift	Drift	drift

Table-1 Emergence of Antigenic Subtypes of Influenza A Virus Associated with Pandemic or Epidemic Disease

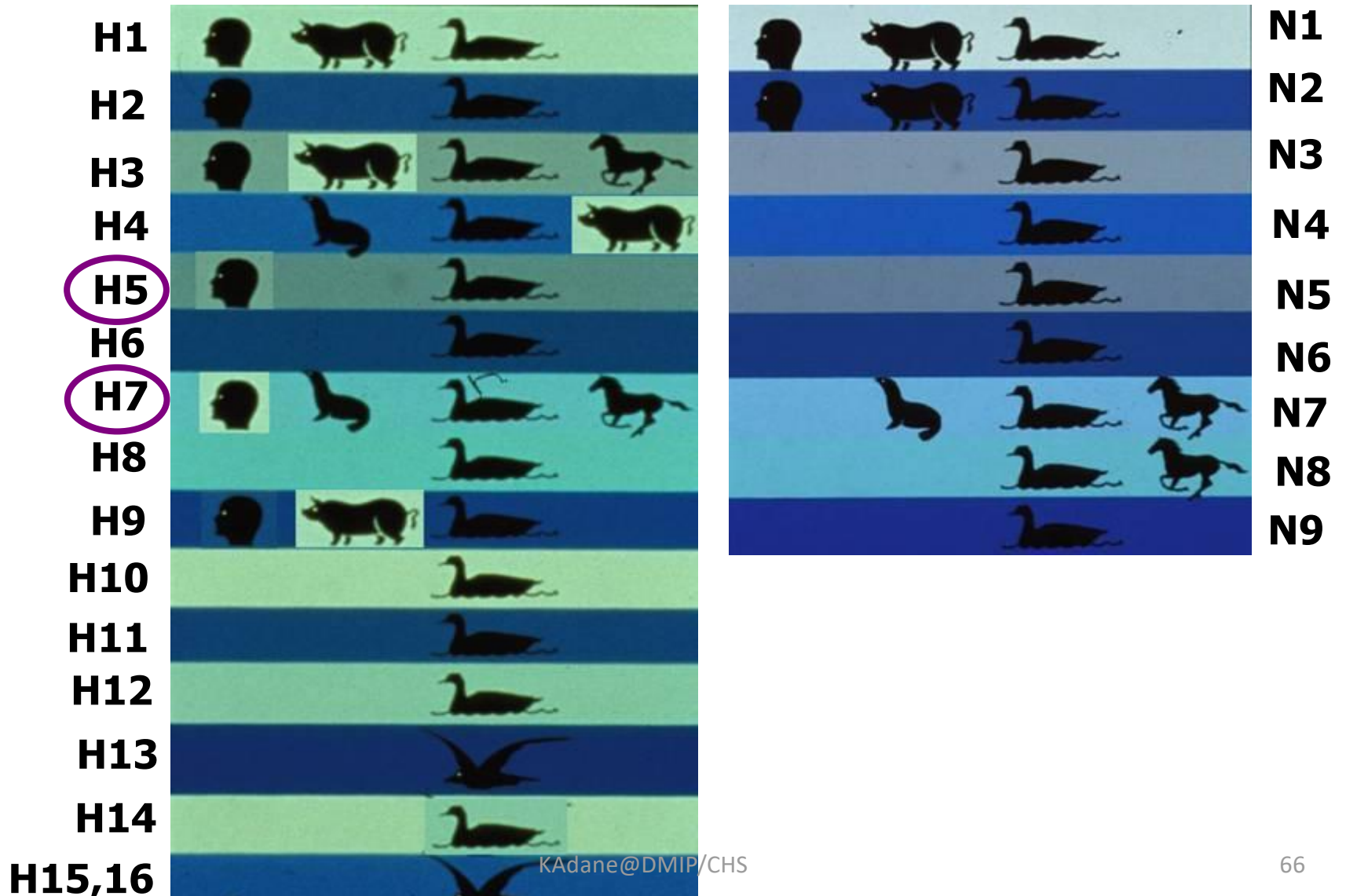
<i>Years</i>	<i>Subtype</i>	<i>Extent of Outbreak</i>
1889–90	H2N8	Severe pandemic
1900–03	H3N8	?Moderate epidemic
1918–19	H1N1 (formerly HswN1)	Severe pandemic
1933–35	H1N1 (formerly H0N1)	Mild epidemic
1946–47	H1N1	Mild epidemic
1957–58	H2N2	Severe pandemic
1968–69	H3N2	Moderate pandemic
1977–78	H1N1	Mild pandemic
2006	H5N1	Bird flu pandemic

Where do “new” HA and NA come from?

- 16 types HA
- 9 types NA
 - All circulate in birds
 - Few circulate in pigs



Species Infected by Influenza A, HA and NA Subtypes



Why Influenza Virus epidemics/pandemics

- Continual evaluation of new strains
 - Antigenic drift
 - Antigenic shift

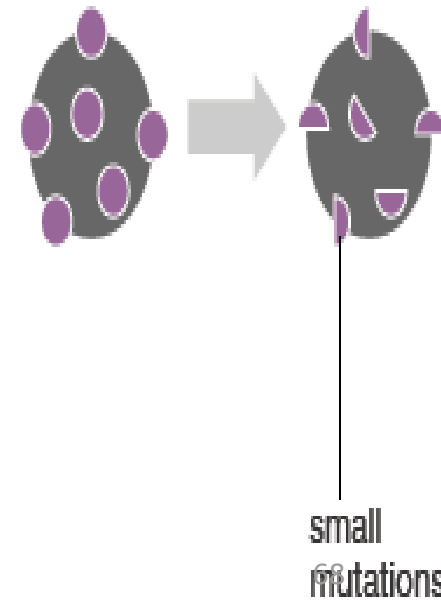
Antigenic Drift & Antigenic Shift

- **Antigenic Drift**

- Minor antigenic changes
- Point mutation
- NEW STRAINS showing minor differences
- TYPE-SPECIFIC IMMUNITY is not entirely protective.
- Infection to continue
- sporadic outbreaks, limited epidemics

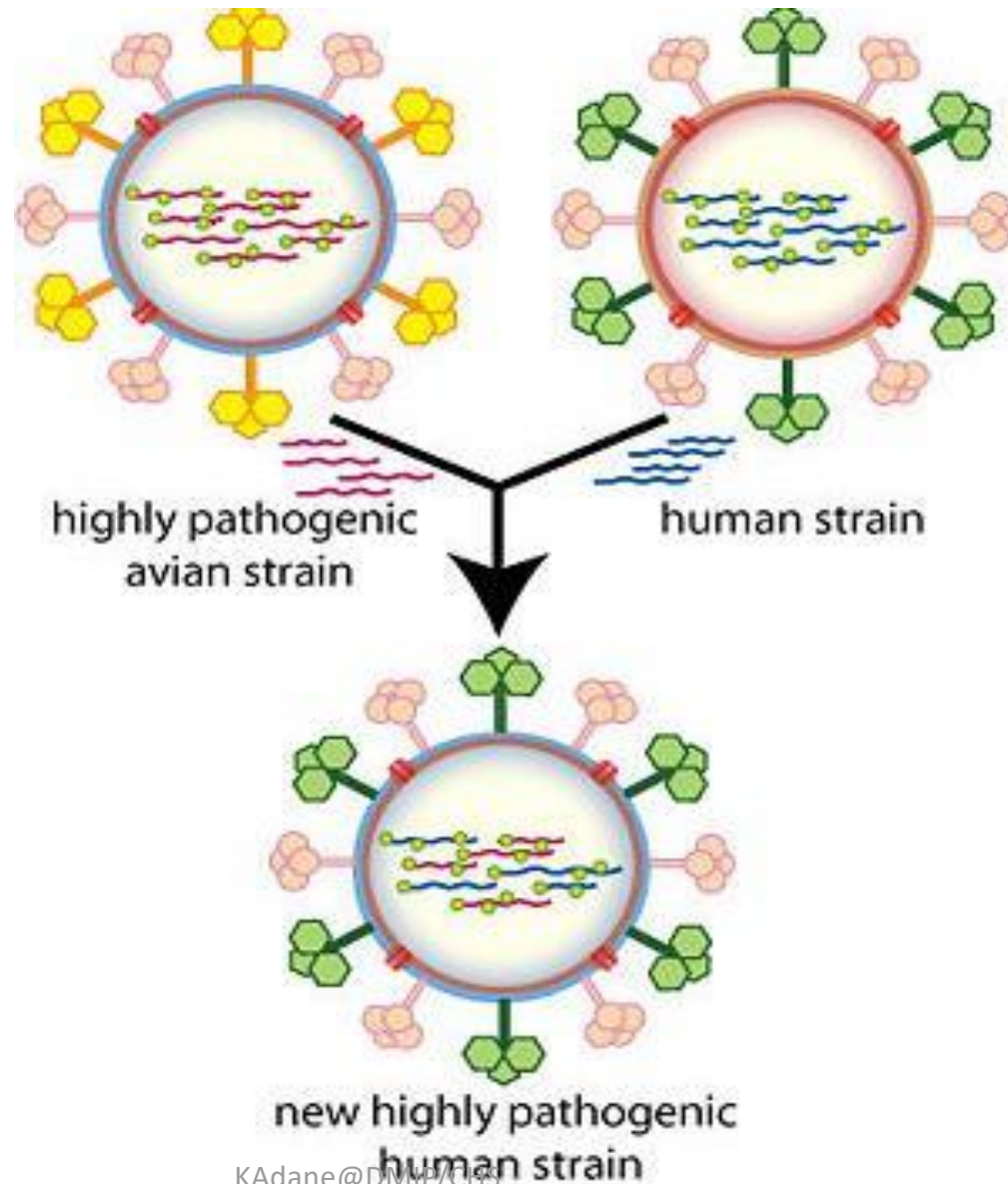
Mutation

Antigenic drift



Antigenic shift

Antigenic shift



Pigs Serve As Mixing Vessels for reassortants

Aquatic birds
and domestic
Poultry



Pigs (vessel)

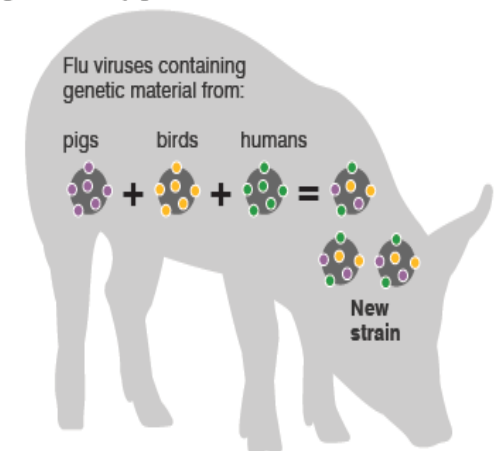


Human
(Human virus)



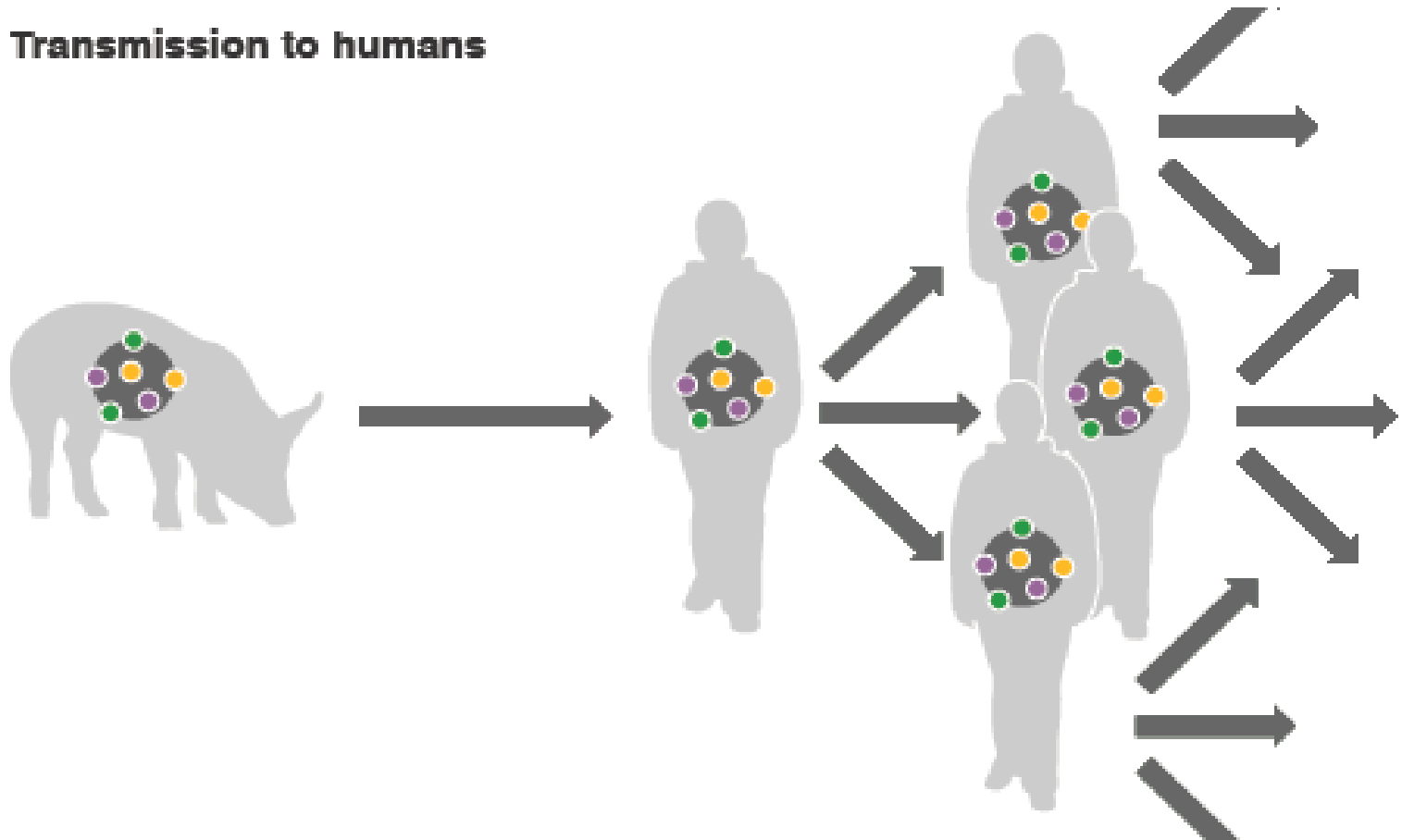
Human virus

Antigenic shift in pigs



Transmission to Humans

Transmission to humans



Human to human Transmission

- AEROSOL
 - 100,000 TO 1,000,000 VIRIONS PER DROPLET
- SHEDDING
- Contact with body secretions



INFLUENZA

- Clinical Findings
- Acute illness
 - **IP 1 -4 DAYS**
- ❖ Flue symptoms
 - FEVER(3-5 days)
 - HEADACHE
 - MYALGIA
 - DRY COUGH
 - RHINITIS
 - ANOREXIA
 - OCULAR SYMPTOMS
 - Recovery 7 to 10 days
 - or COMPLICATION



- **SEVERITY**
 - Elderly
 - Immuno-compromised

Complications

- Primary
 - Influenza virus pneumonia
 - Guillian-Barré syndrome
- Secondary bacterial infections
 - *Streptococcus pneumoniae*
 - *Staphylococcus aureus*
 - *Hemophilus influenzae*



- **Laboratory diagnosis**

- Viral antigens Detection

- RNA detection

- RT-PCR

- Antibody detection

- ELISA, complement fixation

- **Treatment**

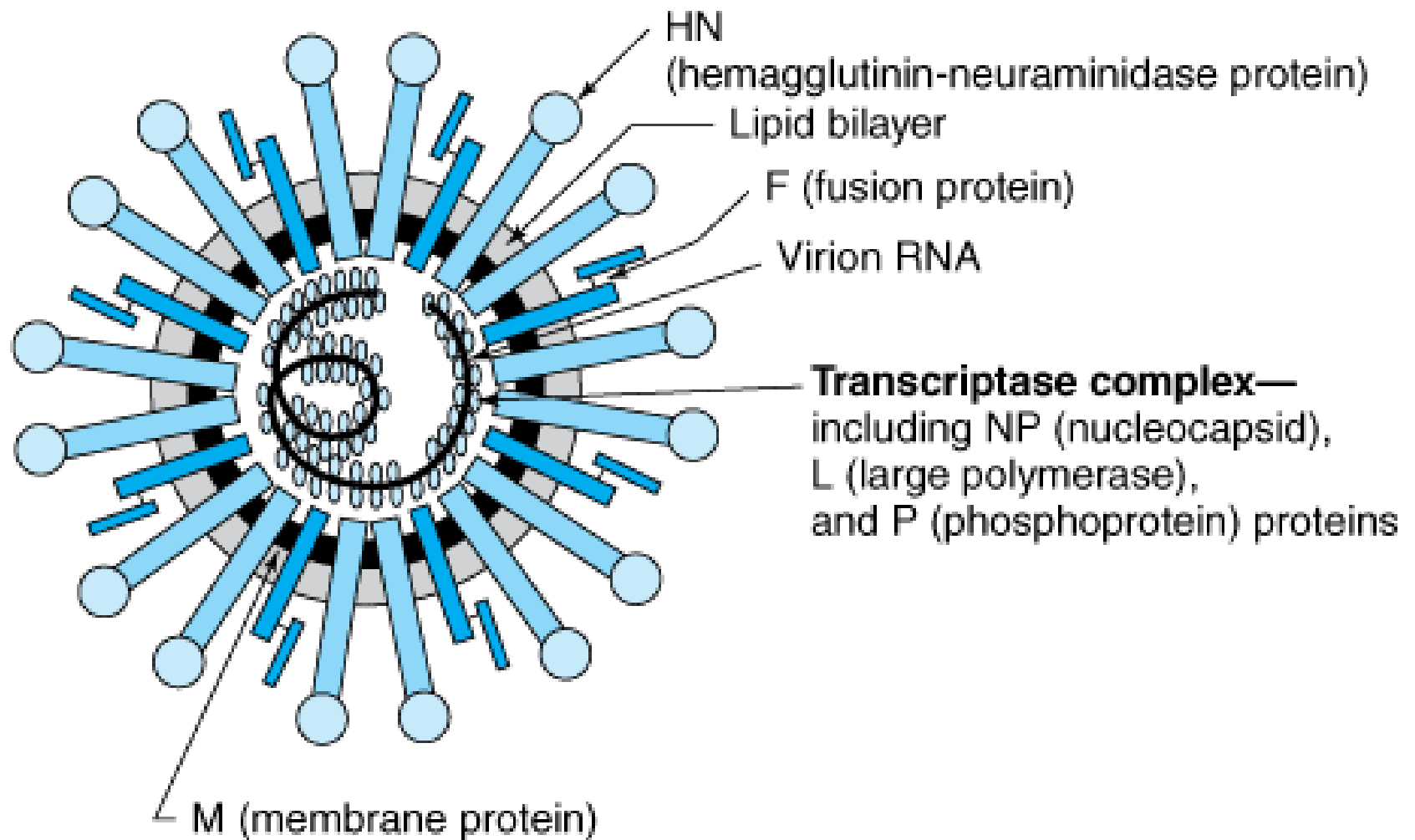
- RIMANTADINE , AMANTADINE , ZANAMIVIR etc

- **Prevention and control**

- Annual vaccination
- Inactivated vaccine (TIV)
- Live Attenuated influenza vaccinem (LAIV)
- Chemoprophylaxis

Measles

- Acute, highly infectious disease characterized by **maculopapular rash, fever and respiratory symptoms**
- Caused by Measles (Rubeola) virus
- Family *Paramyxoviridae*
- Genus-Morbillivirus
 - Enveloped virus
 - Linear, non segmented genome
 - -ss RNA



Transmission

- AEROSOL
- VERY CONTAGIOUS
- (Infectious 4 days before and 4 days after appearance of rash)
- IP : 8 to 12 days, up to 3 weeks in adults



- **Clinical manifestation**

- IP of 7- 11-day illness
- Fever, sneezing, coughing, running nose, conjunctivitis, **Koplik's spots**, photophobia and lymphopenia
- Skin rash



FIGURE 176-1 Koplik's spots, which manifest as white or bluish lesions with an erythematous halo on the buccal mucosa, usually occur in the first 2 days of measles symptoms and may briefly overlap the measles exanthem. The presence of the erythematous halo differentiates Koplik's spots from Fordyce's spots (ectopic sebaceous glands), which occur in the mouths of healthy individuals. (Source: CDC. Photo selected by Dr. Kenneth Kaye.)



- **Laboratory diagnosis**
 - Serology- Antibody detection
 - Isolation of virus- cell culture
- **Treatment**
 - Vitamin A- decrease mortality and morbidity
 - Ribavirin good only experimental not clinically
- **Prevention**
 - Live attenuated vaccine (MMR)
 - Effective- only one strain

- **Thank you!**